11 Chemistry — Unit 2 Topic Test 5: Calorimetry

SOLUTIONS AND MARKING GUIDE — teacher copy; release to students only after the test has been sat

Total 40 marks: Section A 10 × 1 mark (separate answer sheet), Section B 30 marks. Writing time 46 min (40 marks × 1.15 min per mark). Students use the VCE Chemistry Data Book (SLC 25 °C and 100 kPa; V_m = 24.8 L mol^{-1} at SLC; R = 8.31 J mol^{-1} K^{-1}; 1 atm = 101.3 kPa; c(water) = 4.18 J g^{-1} °C^{-1}; T(K) = T(°C) + 273). **One mark per idea or step; no half marks.** Mark units and significant figures **once per part, at the final answer** (3 significant figures expected unless stated; accept 2–4 where the data allow). **Rounding tolerance:** accept a final answer within ±1 in the last significant figure when it comes from correctly rounded intermediate values; each part states its accepted range. **Consequential marking:** where a later part uses an earlier answer, award full marks for correct method applied to the student's earlier value; parts where this applies are flagged *Consequential*. Energy released or absorbed is a positive magnitude. Equations need correct states.

Section A — Multiple-choice answer key (10 marks)

Key: C A B D B A D C A B. Balance: A ×3, B ×3, C ×2, D ×2; no letter more than 1 in a row. Each wrong letter points to one slip or misconception (right-hand column). Difficulty and demand level for each item are shown in italics after the working.

Q	Key	Working for the key	Distractor diagnosis
1	C	The water cools, so it loses energy to the dissolving KCl: the process absorbs energy, so it is endothermic. *[Easy; L1]*	**A** reads a temperature fall as energy released; **B** correct direction but wrong name for the process; **D** correct name but reversed direction of energy transfer
2	A	Polystyrene is a poor conductor of heat and the lid reduces convection, so less energy is lost or gained. *[Easy; L1]*	**B** thinks the cup should absorb energy (that would lower ΔT); **C** over-claim: insulation reduces, but does not eliminate, energy transfer; **D** confuses insulation with a catalyst
3	B	q = mcΔT with c the same for all, so compare m × ΔT: 3000, 4000, 2500, 2400. The 200 g, 20 °C sample needs the most (16.7 kJ). *[Easy; L2]*	**A** chose a middle value without calculating m × ΔT (3000 compared with 4000); **C** chose the largest temperature change only, ignoring mass; **D** chose the largest mass only, ignoring ΔT
4	D	M(C_8H_{18}) = 8(12.0) + 18(1.0) = 114.0 g mol^{-1}; 47.9 × 114.0 = 5.46 × 10^3 kJ for 1.00 mol. *[Medium; L2]*	**A** divided by the molar mass instead of multiplying; **B** treated kJ g^{-1} as kJ mol^{-1}; **C** converted kJ to J (× 1000) as well
5	B	q(water) = 500 × 4.18 × 14.0 = 29 260 J = 29.3 kJ; 29.26 ÷ 60.0 × 100 = 48.8%. *[Medium; L2]*	**A** gave the percentage lost instead of the percentage absorbed; **C** inverted the ratio (energy released ÷ energy absorbed); **D** mixed J and kJ (29.3 J ÷ 60.0 kJ)
6	A	ΔT = highest temperature − temperature before mixing = 27.8 − 19.6 = 8.2 °C. *[Medium; L3]*	**B** used the last reading instead of the highest temperature; **C** used the first reading after mixing as the starting temperature; **D** gave the highest temperature, not the change in temperature
7	D	q = 50.0 × 4.18 × 12.2 = 2550 J = 2.55 kJ; 2.55 ÷ 0.0400 = 63.7 kJ mol^{-1}. *[Medium; L2]*	**A** did not divide by the amount of $CuSO_4$ (this is q in kJ); **B** divided by the mass of $CuSO_4$ (kJ g^{-1}), not the amount; **C** did not convert J to kJ
8	C	Less $CuSO_4$ dissolved, so less energy was released and ΔT is smaller; q is smaller but is still divided by 0.0400 mol, so the calculated value is lower. *[Hard; L4]*	**A** thinks dividing by n cancels the error (but the n used is larger than the amount that dissolved); **B** invents an energy sink; less dissolving means less energy released; **D** thinks ΔT does not depend on the amount that dissolves
9	A	Twice the salt releases twice the energy into twice the water, so ΔT is the same; q and n both double,	**B** doubled the energy released but ignored the doubled mass of water; **C** thinks a larger

		so the energy per mole is the same. *[Hard; L3]*	q means a larger value per mole (n also doubles); **D** doubled the water but ignored the doubled amount of salt
10	B	Relative values (energy reaching the water, the same fraction each time): per gram X 4.60 kJ, Y 3.76 kJ (X higher); per mole X 4.60 × 30.0 = 138 kJ, Y 3.76 × 58.0 = 218 kJ (Y higher). *[Hard; L3]*	**A** assumed the order per gram is also the order per mole; **C** reversed both comparisons; **D** thinks the fuel with the larger molar mass is better per gram too

Section B — Worked solutions and marking scheme (30 marks)

Each part shows the full worked solution, then the bold mark allocation (one mark per idea), then *Consequential* and *Reject* notes. Difficulty and demand level are shown in italics.

Question 1 — Self-heating coffee can (8 marks)

a) (1 mark)

$M(CaO) = 40.1 + 16.0 = 56.1$ g mol^{-1}

$n = 4.20 \div 56.1 = 0.0749$ mol

1 mark 0.0749 mol (accept 0.0748–0.075). *[Easy; L2]*

b) (2 marks)

$\Delta T = 27.8 - 17.0 = 10.8$ °C

$q = mc\Delta T = 100.0 \times 4.18 \times 10.8 = 4514$ J $= 4.51 \times 10^3$ J

1 mark $\Delta T = 10.8$ °C in $q = mc\Delta T$ with $m = 100.0$ g. **1 mark** 4.51×10^3 J (accept 4510–4520 J; unit required). *[Easy; L2]*

Reject: $\Delta T = 27.8$ (11.6 kJ). Mass of CaO + water used without comment (accept only if consistent and stated; the stem says 100.0 g).

c) (2 marks)

$q = 4514$ J $= 4.514$ kJ

Energy per mole $= 4.514 \div 0.07487 = 60.3$ kJ mol^{-1}

1 mark converts J to kJ and divides by n(CaO). **1 mark** 60.3 kJ mol^{-1} (accept 60.2–60.4; positive value; unit required). *[Medium; L2]*

Consequential: On parts a) and b).

Reject: −60.3 kJ mol^{-1} for "energy released" (a sign belongs to ΔH, not to energy released). 6.03×10^4 kJ mol^{-1} (J not converted). 1.07 kJ g^{-1} (per gram).

d) (3 marks)

$q = 200.0 \times 4.18 \times (65.0 - 17.0) = 200.0 \times 4.18 \times 48.0 = 40\,128$ J $= 40.1$ kJ

$n(CaO) = 40.13 \div 60.3 = 0.665$ mol

$m(CaO) = 0.665 \times 56.1 = 37.3$ g

1 mark $q = 4.01 \times 10^4$ J (40.1 kJ). **1 mark** n(CaO) = q ÷ (energy per mole) = 0.665 mol. **1 mark** 37.3 g (accept 37.2–37.4 g). *[Hard; L3]*

Consequential: On the value from part c): m = 40.13 ÷ (c-value) × 56.1.

Reject: $\Delta T = 65.0$ °C (50.6 g). Using 100.0 g of water from the test instead of 200.0 g (18.7 g). Multiplying q by n instead of dividing.

Parts a) to c) are a short chain; d) reverses the direction (energy needed → amount → mass) and is consequential on c).

Question 2 — A gas hot-water service (9 marks)

a) (2 marks)

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ ($H_2O(g)$ also accepted: the stem does not state the state of the water)

1 mark correct formulas and coefficients. **1 mark** all states correct. *[Medium; L2]*

Reject: CO as a product (incomplete combustion). Unbalanced oxygen. A ΔH value is not required and is not penalised if correct.

b) (2 marks)

V = 1.00 m^3 = 1000 L; T = 15.0 + 273 = 288 K; p = 102 kPa

n = pV ÷ RT = (102 × 1000) ÷ (8.31 × 288) = 42.6 mol

1 mark n = pV ÷ RT with V = 1000 L (or 1.00 m^3 with p in Pa) and T = 288 K. **1 mark** 42.6 mol (accept 42.6–42.7). *[Medium; L2]*

Reject: V = 1.00 L (0.0426 mol). n = 1000 ÷ 24.8 = 40.3 mol (V_m at SLC used at 15.0 °C and 102 kPa: 1 mark max). T = 15.0.

c) (3 marks)

Energy released = 42.62 × 890 = 3.79 × 10^4 kJ

q(water) = 135 000 × 4.18 × (60.0 − 14.0) = 135 000 × 4.18 × 46.0 = 2.60 × 10^7 J = 2.60 × 10^4 kJ

Percentage = 2.596 × 10^4 ÷ 3.793 × 10^4 × 100 = 68.4%

1 mark energy released = n × 890 = 3.79 × 10^4 kJ. **1 mark** q(water) = 2.60 × 10^4 kJ (m = 135 000 g). **1 mark** 68.4% (accept 68.3–68.5%). *[Medium; L3]*

Consequential: On n from part b) (e.g. n = 40.3 mol gives 72.3%: full marks here).

Reject: m = 135 g (0.0684%). Inverted ratio (146%). Percentage lost (31.6%).

d) (2 marks)

The average temperature of all the water is lower than the reading at the top, so ΔT used in the calculation is too high.

q(water) is therefore calculated too high, while the energy released by the methane is unchanged, so the calculated percentage is too high.

1 mark the ΔT used is larger than the true average ΔT of the water. **1 mark** so q(water) and the percentage calculated are too high (energy released unchanged). *[Hard; L4]*

Reject: “Lower, because heat is lost.” “Less accurate” with no direction (0). Answers that do not name a direction for the percentage.

Question 3 — Dissolving lithium chloride (6 marks)

a) (1 mark)

Trial 3 (the temperature was recorded before it reached its maximum, so ΔT = 3.6 °C is far too low).

1 mark Trial 3. (A reason is not required.) *[Easy; L3]*

Reject: Trial 2 (lowest initial temperature).

b) (3 marks)

ΔT: trial 1 = 7.2 °C, trial 2 = 7.0 °C; mean ΔT = 7.1 °C

q = 50.0 × 4.18 × 7.1 = 1484 J = 1.48 kJ

n(LiCl) = 1.70 ÷ 42.4 = 0.0401 mol; energy per mole = 1.484 ÷ 0.04009 = 37.0 kJ mol^{-1}

1 mark mean ΔT = 7.1 °C from trials 1 and 2 only. **1 mark** q = 1.48 × 10^3 J. **1 mark** 37.0 kJ mol^{-1} (accept 36.9–37.1; unit required). *[Medium; L3]*

Consequential: On the trials chosen in part a), if the student averaged all three trials (ΔT = 5.9 °C, 30.9 kJ mol^{-1}), award the second and third marks only.

Reject: Using the highest temperature as ΔT. Per gram (0.873 kJ g^{-1}).

c) (2 marks)

With the same ΔT and c, a larger mass (51.7 g instead of 50.0 g) gives a larger q.

The amount of LiCl is unchanged, so the energy per mole is larger (about 38.3 kJ mol^{-1}).

1 mark larger m with the same ΔT and c gives a larger q. **1 mark** n(LiCl) unchanged, so the value per mole is larger. *[Hard; L4]*

Reject: "Smaller, because the LiCl absorbs some of the energy." "No difference." A verdict with no reasoning (0).

Trial 3 is the planted anomaly. Part c) is an evaluation of an assumption; only "larger" scores.

Question 4 — An endothermic demonstration (7 marks)

a) (1 mark)

Endothermic (the water temperature fell: energy was absorbed from the water by the reaction).

1 mark endothermic. *[Easy; L1]*

b) (2 marks)

ΔT = 21.5 − 13.2 = 8.3 °C (a fall)

q = 75.0 × 4.18 × 8.3 = 2602 J = 2.60×10^3 J

1 mark ΔT = 8.3 °C used as a positive change in q = mcΔT with m = 75.0 g. **1 mark** 2.60×10^3 J (accept 2600–2602 J). *[Easy; L2]*

Reject: A negative energy value. ΔT = 13.2 °C. m = 100 g.

c) (3 marks)

$M(Ba(OH)_2 \cdot 8H_2O) = 137.3 + 2(16.0 + 1.0) + 8(18.0) = 315.3$ g mol^{-1}

n = 9.46 ÷ 315.3 = 0.0300 mol

Energy per mole = 2.602 ÷ 0.0300 = 86.7 kJ mol^{-1}

1 mark M = 315.3 g mol^{-1} (including the $8H_2O$). **1 mark** n = 0.0300 mol. **1 mark** 86.7 kJ mol^{-1} (accept 86.6–86.8, or 87 kJ mol^{-1} since ΔT = 8.3 °C has 2 significant figures). *[Medium; L2]*

Consequential: On q from part b).

Reject: M of $Ba(OH)_2$ only, 171.3 g mol^{-1} (n = 0.0552 mol, 47.1 kJ mol^{-1}): loses the first mark only.

d) (1 mark)

So that all of the $Ba(OH)_2 \cdot 8H_2O$ reacts (it is the limiting reactant), so the amount that reacted is known and can be used in part c).

1 mark to make $Ba(OH)_2 \cdot 8H_2O$ the limiting reactant / ensure all of it reacts, so its amount determines the energy absorbed. *[Medium; L3]*

Reject: "To make the reaction faster." "To make it colder" with no reference to all the $Ba(OH)_2 \cdot 8H_2O$ reacting.

Mark summary

Section	Marks	Parts
A — multiple choice	10	10 × 1 mark; key A3 / B3 / C2 / D2
B Q1 — Self-heating coffee can	8	a 1, b 2, c 2, d 3
B Q2 — A gas hot-water service	9	a 2, b 2, c 3, d 2
B Q3 — Dissolving lithium chloride	6	a 1, b 3, c 2
B Q4 — An endothermic demonstration	7	a 1, b 2, c 3, d 1
Total	40	40 marks × 1.15 min = 46 min writing time

Difficulty tally (marks)

Target for every topic test: 25% easy (10 marks), 50% medium (20 marks), 25% hard (10 marks).

	Easy	Medium	Hard	Total
Section A (MC)	3	4	3	10
Section B (short answer)	7	16	7	30
Whole test (marks)	10	20	10	40
Share of marks	25%	50%	25%	100%

Difficulty and demand by item

Item	Marks	Difficulty	Demand
A1	1	Easy	L1
A2	1	Easy	L1
A3	1	Easy	L2
A4	1	Medium	L2
A5	1	Medium	L2
A6	1	Medium	L3
A7	1	Medium	L2
A8	1	Hard	L4
A9	1	Hard	L3
A10	1	Hard	L3
B1a	1	Easy	L2
B1b	2	Easy	L2
B1c	2	Medium	L2
B1d	3	Hard	L3
B2a	2	Medium	L2
B2b	2	Medium	L2
B2c	3	Medium	L3
B2d	2	Hard	L4
B3a	1	Easy	L3
B3b	3	Medium	L3
B3c	2	Hard	L4
B4a	1	Easy	L1
B4b	2	Easy	L2
B4c	3	Medium	L2
B4d	1	Medium	L3

Cognitive demand

Demand level	Marks	Share
L1 Recall / identify	3	8%
L2 Routine procedure	18	45%
L3 Interpret / explain	14	35%
L4 Evaluate / error direction	5	12%

Study design coverage

Key knowledge (Unit 2)	Where assessed
Specific heat capacity of water; $q = mc\Delta T$ (units, ΔT, J ↔ kJ)	A3, A6; B1b, B2c, B3b, B4b
Exothermic and endothermic reactions from the sign of ΔT	A1; B4a
Solution calorimetry: energy per mole	A7; B1c, B3b, B4c
Combustion calorimetry heating water: energy per gram / per mole	A4, A10; B2
Percentage of energy transferred	A5; B2c
Heat losses, assumptions and error direction	A2, A8, A9; B2d, B3a, B3c, B4d
Linked skills: $pV = nRT$, limiting reactant, mass ↔ mol	B1d, B2b, B4d